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Linguistic Markers of Subclinical Cerebrovascular Compromise: A Theoretical Framework and Proposed Validation Study Using Neural Oscillation Signature Theory

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The author is the founder and president of Beverly Index LLC, a company developing commercial applications based on the theoretical framework presented in this paper. This potential conflict of interest is disclosed in accordance with APA ethical standards. The theoretical framework and proposed study design presented herein are offered for scientific scrutiny and collaborative development by the research community.

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Abstract

Cerebrovascular disease remains a leading cause of death and disability worldwide, yet current detection methods—neuroimaging, coagulation panels, and clinical symptom assessment—are poorly suited to early identification of the subclinical vascular compromise that precedes major cerebrovascular events by months or years. This paper proposes a novel detection pathway based on Neural Oscillation Signature Theory (NOST; Beverly, 2026b), a theoretical framework positing that neural oscillatory states leave detectable signatures in linguistic output.

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Because cerebrovascular pathology—including chronic small vessel disease, subclinical microthrombi, transient ischemic events, and progressive hypoperfusion—alters cerebral neural oscillatory activity in well-documented ways, NOST predicts that these vascular changes should produce detectable shifts in the statistical properties of language before conventional clinical assessment identifies cognitive decline. This paper articulates the theoretical chain linking blood pathology to oscillatory disruption to linguistic signature change, reviews the supporting empirical literature from vascular neurology, clinical electroencephalography, and psycholinguistics, specifies the predicted linguistic markers of subclinical cerebrovascular compromise, and proposes a prospective longitudinal validation study design. If confirmed, this approach would establish language-based monitoring as an early warning system for cerebrovascular events—a scalable, non-invasive complement to existing detection methods with particular relevance for populations at elevated vascular risk.

Keywords: cerebrovascular disease, subclinical detection, neural oscillations, linguistic markers, NOST, NOSS, vascular cognitive impairment, early warning, stroke prevention, small vessel disease

Linguistic Markers of Subclinical Cerebrovascular Compromise:

A Theoretical Framework and Proposed Validation Study

Stroke is the second leading cause of death worldwide and a primary cause of acquired disability in adults (Feigin et al., 2022). Despite advances in acute treatment—particularly the expansion of thrombectomy and thrombolytic therapy windows—the fundamental clinical challenge remains early detection: the majority of cerebrovascular events occur without adequate prior warning, in patients whose subclinical vascular compromise was not identified by existing assessment methods (Wardlaw et al., 2019). The economic burden of stroke in the United States

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alone exceeds $50 billion annually (Virani et al., 2021), a figure that does not capture the full cost of post-stroke disability, caregiver burden, and lost productivity.

This paper proposes that a significant and underexploited early detection pathway exists in the linguistic output of individuals with subclinical cerebrovascular compromise. The proposal rests on Neural Oscillation Signature Theory (NOST; Beverly, 2026b), a theoretical framework establishing that neural oscillatory states systematically shape language production in ways that leave detectable statistical signatures in spoken and written text. Because cerebrovascular pathology directly alters neural oscillatory activity—through ischemia, hypoperfusion, microhemorrhage, or diffuse small vessel disease—NOST predicts that these vascular changes should produce measurable shifts in linguistic features before cognitive decline reaches the threshold of conventional clinical detection.

The present paper is theoretical. It articulates the detection chain linking vascular pathology to oscillatory disruption to linguistic change, reviews the empirical evidence supporting each link, specifies the predicted linguistic markers, and proposes a validation study designed to test the framework's predictions. No empirical results are presented. The framework is offered for scientific scrutiny and collaborative development.

The Detection Chain: From Blood Pathology to Linguistic Signature

The proposed detection pathway involves four sequential links, each supported by established empirical evidence but not previously integrated into a unified early detection framework.

Link 1: Blood Pathology Alters Cerebral Perfusion

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The first link is the most straightforward: hematological and vascular pathology compromises cerebral blood flow. This occurs through multiple mechanisms operating across different timescales and severity levels.

Acute mechanisms include large-vessel occlusion (ischemic stroke), intracerebral hemorrhage, and pulmonary embolism producing systemic hypoxia. These produce immediate, often catastrophic disruption of cerebral perfusion and are typically clinically obvious. They are not the primary target of the present framework, though NOST-based monitoring could contribute to their detection in ambiguous cases.

Subacute mechanisms include transient ischemic attack (TIA), which produces temporary focal neurological deficit that resolves within 24 hours but signals elevated stroke risk (Rothwell et al., 2005). Critically, many TIAs go unrecognized by patients, particularly those producing subtle cognitive rather than motor symptoms (Coutts et al., 2012). These unrecognized transient events represent a high-value detection target.

Chronic mechanisms include cerebral small vessel disease (SVD), the most prevalent cerebrovascular condition and the primary substrate of the proposed detection framework. SVD encompasses arteriosclerosis of small penetrating arteries and arterioles, leading to lacunar infarcts, white matter hyperintensities, microbleeds, and diffuse white matter damage (Wardlaw et al., 2019). Hypercoagulable states—whether genetic (Factor V Leiden, prothrombin G20210A mutation, antiphospholipid syndrome) or acquired—increase the frequency of subclinical microthrombotic events that contribute to cumulative small vessel damage. Chronic hypertension, diabetes mellitus, and hyperlipidemia further accelerate this process. The critical clinical feature of SVD is that it progresses for years in a largely subclinical fashion, producing

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gradual cognitive decline that patients and clinicians frequently attribute to normal aging until a major cerebrovascular event occurs (Debette & Markus, 2010).

Link 2: Impaired Cerebral Perfusion Disrupts Neural Oscillatory Activity

The second link draws on the well-established electroencephalographic literature documenting oscillatory changes associated with cerebrovascular compromise. Neural oscillatory activity is exquisitely sensitive to perfusion changes because oscillatory synchrony depends on metabolically demanding synaptic processes that are among the first to degrade under ischemic or hypoxic conditions.

The documented oscillatory consequences of cerebrovascular pathology include: increased delta-band (0.5–4 Hz) power during wakefulness, reflecting cortical slowing in ischemic regions (Finnigan & van Putten, 2013); reduced alpha-band (8–12 Hz) power and slowed peak alpha frequency, reflecting diffuse cortical compromise (Schleiger et al., 2014); disrupted cross-frequency coupling between theta and gamma bands, reflecting impaired coordination between hippocampal memory systems and cortical processing networks (Moretti et al., 2012); and reduced beta-band modulation during cognitive tasks, reflecting compromised executive function circuits (Babiloni et al., 2010). These changes are dose-dependent: more severe perfusion compromise produces more pronounced oscillatory disruption, creating a graded signal that should be detectable at subclinical levels if measurement sensitivity is sufficient.

Quantitative EEG studies have demonstrated that these oscillatory changes are present in patients with documented SVD even when standard neuropsychological testing remains within normal limits (Moretti et al., 2012; Babiloni et al., 2010). This finding is critical for the present framework: it establishes that oscillatory disruption precedes detectable cognitive decline on

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conventional assessment, providing the temporal gap that NOST-based linguistic monitoring could exploit for early detection.

Link 3: Disrupted Oscillatory Activity Alters Language Production

The third link is the core claim of NOST (Beverly, 2026b): neural oscillatory patterns systematically shape language production at multiple levels—lexical selection, syntactic construction, semantic organization, temporal structure, and prosodic characteristics—such that the statistical properties of linguistic output carry extractable information about the oscillatory configuration that generated them. The companion papers establishing this framework (Beverly, 2026a, 2026b, 2026c) provide the complete theoretical development and are not recapitulated here.

For the present application, the relevant predictions are specific: the oscillatory disruptions documented in cerebrovascular disease should produce corresponding linguistic changes organized by the oscillatory bands affected. Delta-band slowing should reduce global narrative coherence and thematic integration. Alpha-band compromise should alter attentional markers, producing reduced information selection efficiency and increased susceptibility to irrelevant content intrusion. Theta-band disruption should impair memory-dependent linguistic features, including autobiographical detail and temporal reference accuracy. Beta-band compromise should reduce syntactic complexity and sequential planning in discourse. Gamma-band disruption, if present, should affect semantic coherence and perceptual description fidelity.

Critically, the vascular cognitive impairment literature identifies a characteristic cognitive profile that differs from other causes of cognitive decline: SVD preferentially affects processing speed, executive function, and attentional control, while relatively preserving episodic memory in early stages (O'Brien et al., 2003). This profile predicts a distinctive NOSS activation pattern:

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primary elevation in S4 (Neurocognitive) with executive-predominant features, distinguishable from the memory-predominant pattern expected in S8 (Neurodegenerative, Alzheimer's-type) and from the focal, lateralized pattern expected in S7 (TBI) following acute stroke. The specificity of the predicted vascular linguistic signature, if confirmed, would provide not merely early detection of cognitive decline but differential diagnostic information about its likely etiology.

Link 4: Computational Extraction Enables Subclinical Detection

The fourth link is methodological: modern computational linguistic methods can detect statistical regularities in language that are invisible to human observation, including changes too subtle for the speaker to notice or the clinician to identify through standard interview. This capacity for detecting subthreshold changes is essential because the target population—individuals with subclinical cerebrovascular compromise—by definition presents without obvious cognitive symptoms.

The relevant computational methods include: automated measurement of syntactic complexity through parse tree analysis, which can detect reductions in clause embedding depth and sentence length before they reach the threshold of clinical notice (Fraser et al., 2016); semantic coherence analysis through embedding similarity, which quantifies the degree to which discourse maintains thematic integration across segments (Elvevåg et al., 2007); information density measurement, which tracks the efficiency of information packaging in discourse (Jaeger, 2010); and temporal reference analysis, which quantifies the accuracy and specificity of time-related language. Each of these methods can detect graded changes along a continuous dimension, avoiding the binary detection threshold that limits conventional clinical assessment.

Predicted Linguistic Signatures of Subclinical Cerebrovascular Compromise

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Based on the detection chain articulated above, the following specific linguistic markers are predicted to be affected by subclinical cerebrovascular compromise. These predictions are organized by the clinical mechanism producing them and are designed to be testable in the proposed validation study.

Vascular Executive Signature

Cerebral small vessel disease preferentially affects frontal-subcortical circuits supporting executive function (O'Brien et al., 2003). The predicted linguistic consequences include: reduced syntactic complexity (fewer embedded clauses, shorter mean sentence length, decreased use of subordinate constructions), diminished discourse planning (reduced use of prospective language, fewer discourse markers signaling upcoming content organization), impaired cognitive control markers (increased topic drift, reduced ability to maintain coherent narrative across extended discourse), and slowed information processing reflected in increased pause frequency and duration in spoken language samples.

Vascular Processing Speed Signature

Processing speed decline is the earliest and most sensitive cognitive marker of SVD (Prins & Scheltens, 2015). The predicted linguistic consequences include: reduced information density (fewer propositions per utterance), increased filler usage and hesitation markers, reduced word-finding efficiency (increased circumlocution, tip-of-tongue phenomena reflected in self-correction patterns), and decreased overall output rate in time-constrained language samples. These features should be detectable in both spoken and written samples, with spoken samples providing the additional dimension of temporal dynamics.

Vascular Attentional Signature

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Impaired attentional control in SVD reflects disrupted alpha-band gating and compromised frontal-parietal attention networks. The predicted linguistic consequences include: increased susceptibility to tangential content (reduced relevance maintenance), altered information selection (inclusion of contextually inappropriate detail alongside omission of relevant information), and reduced coherence maintenance across extended narrative samples. This signature overlaps with but is distinguishable from the attentional signature predicted for S2 (Anxiety) in the NOSS system, because vascular attentional impairment lacks the threat-related content bias and hypervigilance markers characteristic of anxiety-driven attentional disruption.

Transient Ischemic Signature

Unrecognized transient ischemic events are predicted to produce a distinctive temporal pattern: acute, relatively focal linguistic disruption followed by partial or complete resolution over hours to days. The Psychological Lense-Thirring Precession model (Beverly, 2026b) predicts that the perturbation should follow damped oscillatory dynamics, with initial linguistic disruption followed by recovery that may include residual subtle deficits detectable through sensitive computational analysis. Serial monitoring—repeated language sampling at regular intervals—would be required to detect these transient perturbations. The clinical value of detecting unrecognized TIAs is substantial: TIA is the single strongest predictor of imminent major stroke, with 90-day stroke risk exceeding 10% in some populations (Rothwell et al., 2005).

Longitudinal Trajectory Signature

Perhaps the most diagnostically valuable prediction concerns trajectory rather than snapshot: NOST predicts that subclinical cerebrovascular compromise should produce a measurable downward trend in specific linguistic features over serial assessments. A patient whose baseline executive-linguistic profile was established during healthy function should show

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statistically detectable degradation in syntactic complexity, processing efficiency, and discourse planning over months to years of progressive SVD—degradation that precedes the clinical threshold for diagnosis of vascular cognitive impairment. The trajectory signature is predicted to be more sensitive than any single-point assessment because it exploits the individual's own baseline as the comparison standard, eliminating inter-individual variability in baseline linguistic ability.

Proposed Validation Study

Study Design

The proposed study is a prospective longitudinal cohort design with a 36-month follow-up period. The design is specifically structured to test whether NOST-based linguistic monitoring detects subclinical cerebrovascular compromise before conventional clinical assessment, and whether linguistic trajectory changes predict subsequent cerebrovascular events.

Participants

Three cohorts would be recruited: (1) a high-risk cohort (n = 200) comprising adults aged 55–75 with at least two established cerebrovascular risk factors (hypertension, diabetes mellitus, hyperlipidemia, atrial fibrillation, known hypercoagulable state, prior TIA, or family history of stroke) but no history of clinical stroke or diagnosed cognitive impairment; (2) an age-matched control cohort (n = 100) with no more than one cerebrovascular risk factor; and (3) an existing SVD cohort (n = 100) with MRI-confirmed cerebral small vessel disease but without clinical dementia, serving as a positive reference group for the predicted vascular linguistic signature.

Baseline Assessment

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All participants would undergo comprehensive baseline evaluation including: structural MRI with white matter hyperintensity volumetric quantification and microbleed assessment; quantitative EEG with spectral power analysis across standard frequency bands; comprehensive neuropsychological testing battery including measures of processing speed (Symbol Digit Modalities Test), executive function (Trail Making Test B, Stroop), memory (Rey Auditory Verbal Learning Test), and attention (Continuous Performance Test); complete coagulation panel including fibrinogen, D-dimer, antiphospholipid antibodies, Factor V Leiden screening, and prothrombin gene mutation analysis; and standardized cardiometabolic risk assessment.

Linguistic Data Collection

Participants would provide standardized narrative samples at monthly intervals for 36 months. Each monthly session would include: a semi-structured autobiographical narrative (5-minute spoken account of a recent personally experienced event), a standardized picture description task (Cookie Theft picture from the Boston Diagnostic Aphasia Examination), a written narrative sample (typed account of a dream from the preceding week, following REMIEL Engine collection protocols; Beverly, 2026d), and an optional daily brief voice sample (60-second spoken diary entry via smartphone application) for participants willing to provide higher-frequency data.

All spoken samples would be transcribed using automated speech recognition with manual verification. Both spoken and written samples would be processed through a comprehensive computational linguistic analysis pipeline extracting features across five domains: lexical (vocabulary diversity, word frequency distribution, pronoun patterns, emotional valence), syntactic (mean sentence length, clause embedding depth, subordination ratio, syntactic similarity), semantic (embedding-based coherence, propositional density, referential clarity,

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thematic integration), temporal (processing time, pause distribution, speech rate, hesitation frequency), and prosodic (pitch variability, voice quality measures, stress patterns, intonation contour regularity).

Follow-Up Assessment

Full neuropsychological, neuroimaging, and vascular assessment would be repeated at 12, 24, and 36 months, providing criterion measures against which linguistic trajectory data can be evaluated. Interim clinical events—including diagnosed TIA, stroke, newly identified SVD on imaging, initiation of anticoagulation therapy, or new diagnosis of vascular cognitive impairment—would be documented as they occur.

Primary Analyses

The proposed analyses address four sequential questions.

First, cross-sectional discrimination: at baseline, do participants with confirmed SVD (Cohort 3) exhibit the predicted vascular linguistic signature—executive-predominant linguistic features distinguishable from age-matched controls (Cohort 2)? This analysis establishes whether the predicted vascular signature exists.

Second, subclinical detection: within the high-risk cohort (Cohort 1), do participants who subsequently develop MRI-evident SVD, diagnosed vascular cognitive impairment, or cerebrovascular events during the 36-month follow-up show detectable linguistic trajectory changes before the clinical event? This analysis tests the early detection hypothesis.

Third, temporal precedence: among participants with documented clinical events, how far in advance do statistically significant linguistic trajectory changes appear? This analysis quantifies the potential early warning window.

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Fourth, differential signature: can the linguistic trajectory pattern associated with subclinical cerebrovascular compromise be distinguished from age-related cognitive decline in the control cohort? This analysis tests specificity—the claim that vascular cognitive impairment produces a recognizably different linguistic signature than normal aging.

Power Considerations

The proposed sample sizes are estimated based on the effect sizes reported in the existing literature on linguistic markers of cognitive decline. Fraser et al. (2016) reported classification accuracies exceeding 80% for distinguishing Alzheimer's disease from healthy controls using linguistic features alone. The present application targets a more subtle distinction—subclinical vascular compromise versus healthy aging—and therefore requires larger samples and higher-frequency longitudinal measurement to achieve adequate power. The high-risk cohort of 200 participants with monthly sampling provides approximately 7,200 data points over 36 months, sufficient for mixed-effects modeling of individual trajectories with reasonable statistical power for detecting small-to-medium effect sizes.

Clinical Implications

If the predicted linguistic signatures of subclinical cerebrovascular compromise are confirmed, the clinical implications are substantial.

Early Warning System for Cerebrovascular Events

The most immediate application would be the development of a language-based early warning system for individuals at elevated cerebrovascular risk. Regular linguistic monitoring—potentially through as little as a weekly smartphone voice sample or written dream report—could provide continuous tracking of the cognitive consequences of progressive vascular compromise,

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triggering clinical alerts when trajectory changes exceed statistically defined thresholds. This monitoring would complement rather than replace existing risk assessment methods (blood pressure monitoring, lipid panels, coagulation studies), providing a functional readout of whether vascular risk factors are producing measurable cognitive impact.

Treatment Response Monitoring

NOST's Treatment Sensitivity Principle (Beverly, 2026b, Proposition 5) predicts that linguistic signatures should change in response to effective treatment. If confirmed, linguistic monitoring could provide objective assessment of whether anticoagulation therapy, antihypertensive treatment, statin initiation, or other interventions are producing measurable cognitive benefit—information that is currently available only through neuropsychological retesting at multi-month intervals. More frequent linguistic monitoring could detect treatment response (or failure) on a timeline of weeks rather than months.

Population-Level Screening

Because linguistic data collection requires no specialized equipment, no clinical visit, and no trained administrator, NOST-based vascular monitoring is inherently scalable. A validated linguistic screening tool could be deployed through primary care networks, integrated into telemedicine platforms, or offered as a consumer health monitoring application—reaching populations that lack access to neuroimaging, specialist vascular assessment, or comprehensive neuropsychological evaluation. The public health potential for stroke prevention through early detection of subclinical cerebrovascular compromise is significant.

Relationship to Existing Detection Approaches

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The proposed framework does not compete with existing cerebrovascular detection methods. Rather, it addresses a specific gap in the current detection landscape: the interval between the onset of subclinical vascular compromise and its detection by conventional methods.

Neuroimaging (MRI) provides definitive evidence of cerebral small vessel disease through identification of white matter hyperintensities, lacunar infarcts, and microbleeds. However, MRI is expensive, requires specialized facilities, and is not performed routinely in asymptomatic patients. The proposed framework could serve as a screening tool that identifies candidates for neuroimaging—reducing unnecessary imaging while improving detection of subclinical disease.

Coagulation studies and vascular biomarkers identify systemic risk but do not directly measure cerebral impact. A patient with a hypercoagulable state may or may not have experienced cerebral microthrombotic events; coagulation panels cannot distinguish between these scenarios. Linguistic monitoring measures the functional cerebral consequence of whatever vascular pathology is present, regardless of its specific mechanism.

Standard neuropsychological testing detects cognitive impairment but requires clinical visits, trained administrators, and typically has limited sensitivity to the earliest stages of vascular cognitive decline, particularly when patients are educated and have high cognitive reserve (Stern, 2009). Linguistic monitoring offers higher-frequency sampling, reduced practice effects (because narrative generation tasks have no fixed correct answers), and sensitivity to within-person change that leverages the individual's own baseline rather than population norms.

Limitations and Caveats

Several significant limitations must be acknowledged.

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First, the entire framework is theoretical. No empirical data currently exist testing whether NOST-based linguistic monitoring detects subclinical cerebrovascular compromise. The detection chain is logically coherent and each individual link is supported by existing evidence, but the integrated system has not been validated. The proposed study is designed to provide this validation; until it is conducted, the framework remains speculative.

Second, NOST detects the cognitive consequence of vascular pathology, not the vascular pathology itself. A linguistic trajectory change consistent with the predicted vascular signature could be produced by any condition affecting the same neural circuits—including non-vascular causes of executive dysfunction, medication effects, depression, or other comorbidities. The framework provides detection and differential characterization, not definitive etiological diagnosis. Medical investigation following a linguistic alert would remain essential.

Third, the proposed vascular linguistic signature overlaps with signatures of normal aging. Disentangling accelerated vascular decline from the upper end of normal age-related cognitive change requires the longitudinal trajectory analysis proposed here, but the boundary between these categories is inherently gradual rather than sharp. False positive rates in older populations may be substantial and must be empirically quantified.

Fourth, individual differences in baseline linguistic ability, education, bilingualism, and habitual communication style introduce variability that could obscure vascular-related changes. The proposed study design addresses this through within-person trajectory analysis, but the minimum sampling frequency required for reliable individual-level detection remains an open empirical question.

Fifth, the author's commercial interest in NOST-based applications creates a potential bias that the scientific community should weigh. Independent replication and critique of both the

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theoretical framework and any empirical findings will be essential. The proposed study would benefit from collaboration with independent vascular neurology research groups without commercial ties to the technology.

Conclusion

This paper proposes that the linguistic output of individuals with subclinical cerebrovascular compromise carries detectable signatures of the underlying vascular pathology—signatures mediated by the effects of impaired cerebral perfusion on neural oscillatory activity and, consequently, on the cognitive processes that shape language production. The theoretical chain linking blood pathology to oscillatory disruption to linguistic change is supported at each link by established empirical evidence, and the predicted vascular linguistic signature—executive-predominant, processing-speed-sensitive, and trajectory-dependent—is specific enough to be empirically tested and potentially falsified.

The proposed validation study provides a structured research program for testing these predictions. If confirmed, the framework would establish language-based monitoring as a scalable, non-invasive early warning system for cerebrovascular events—addressing a critical gap in the current detection landscape between the onset of subclinical vascular compromise and its identification by conventional clinical methods.

The clinical stakes are substantial. Cerebrovascular disease is among the most preventable causes of catastrophic neurological injury. Early detection of subclinical vascular compromise—months or years before major stroke—would enable timely intervention with anticoagulation, antihypertensive therapy, lifestyle modification, and enhanced surveillance. If the words we speak carry the signatures of the vascular processes affecting our brains, then

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listening carefully—computationally, systematically, and longitudinally—may prove to be among the most consequential advances in stroke prevention.

The scientific community's engagement—through critique, collaboration, and independent investigation—is both essential and welcomed.

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